THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bag contains 11 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{16}{25}$. The value of x is: 1

(a) 5 **(b)** 6 **(c)** 10 **(d)** 4

2. A bag contains 13 red and 7 blue balls. 7 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{7}{27}$ **(b)** $\frac{7}{9}$ **(c)** $\frac{20}{27}$ **(d)** $\frac{19}{27}$

3. A box contains 80 tickets numbered 1 to 80. One ticket is drawn at random. The probability that the number drawn is a multiple of 3 or 5 is: 1

(a) $\frac{9}{20}$ **(b)** $\frac{19}{40}$ **(c)** $\frac{43}{80}$ **(d)** $\frac{37}{80}$

4. A lot of 180 ball pens contains 27 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{38}{45}$ **(b)** $\frac{77}{90}$ **(c)** $\frac{17}{20}$ **(d)** $\frac{3}{20}$

5. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that both dice show the same number (a doublet) is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{7}{36}$ **(c)** $\frac{5}{6}$ **(d)** $\frac{5}{36}$

6. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that both numbers are odd is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{5}{18}$ **(c)** $\frac{2}{9}$ **(d)** $\frac{3}{4}$

7. A bag contains 14 red balls and x blue balls. The probability of drawing a blue ball is 5 times the probability of drawing a red ball. The value of x is: 1

(a) 69 **(b)** 84 **(c)** 5 **(d)** 70

8. A bag contains 5 red and 13 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{12}{25}$. The value of x is: 1

(a) 14 **(b)** 6 **(c)** 7 **(d)** 8

9. **Assertion (A):** The probability of an event can be 1.5. 1

Reason (R): For any event E, $0 \le P(E) \le 1$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** When two coins are tossed together, the probability of getting at least one head is $\frac{3}{4}$. 1

Reason (R): The sum of the probabilities of all the elementary events of an experiment is 1.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A bag contains 16 red balls and x blue balls. The probability of drawing a blue ball is 5 times the probability of drawing a red ball. Find the value of x. 2

12. A bag contains 12 red and 8 blue balls. 8 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 2

13. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is 12. 2

14. A bag contains 20 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. Find the value of x. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that at least one die shows 6. 3

16. A bag contains 17 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{8}{11}$. Find the value of x. 3

17. A box contains 96 tickets numbered 1 to 96. One ticket is drawn at random. Find the probability that the number drawn is a multiple of 3 or 5. 3

OR

Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is 3 or 11. 3

18. A bag contains 6 red and 14 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{8}{15}$. Find the value of x. 3

OR

A box contains 72 tickets numbered 1 to 72. One ticket is drawn at random. Find the probability that the number drawn is a multiple of 2 but not of 4. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two fair dice are thrown together in a board game. 5
(i) How many outcomes are possible in all?
(ii) Find the probability that the sum of the numbers is 8.
(iii) Find the probability of getting a doublet.
(iv) Find the probability that the sum is a prime number, showing the count of favourable outcomes.

20. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum is at least 10,
(ii) the sum is 9,
(iii) the larger number is 4.

OR

A bag contains 11 red and 25 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{16}{41}$. Find the value of x. 5

END OF QUESTION PAPER — Total: 40 Marks